

Akai Force × FL Studio

Live Control script: User Manual



Version 1.0.0

github.com/Pyrodrifter/akai-force-fl-studio

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1. Overview

This script turns an **Akai Force** into a full controller for **FL Studio**, the same way the Force works with Ableton Live: its touchscreen shows your tracks, clips and plugin parameters, the 8×8 pads launch clips, sequence steps and play notes, and the knob screens show names and values.

How it works

The Force has a **Live Control** mode made for Ableton Live. In that mode it talks plain MIDI over the network through Akai's **Network MIDI driver**, on a port called `Akai Network - DAW Control`. This project re-implements the other side of that conversation as an FL Studio **MIDI controller script**, so FL Studio takes Ableton's place. Nothing else runs in the background: no server, no VST, no bridge app.



The Force draws its own screens. The script only sends names, colours and values, and reacts to your pads, knobs, buttons and touches.

What you get

- **Seven pad modes:** PERFORM (clip launching), PATTERNS, STEPS (step sequencer with per-step editing), KEYS (with scales), DRUMS, CHANNELS and SONG (jump to any bar).
- **Mixer on the Force:** 8 mixer inserts at a time with names, colours, volume, pan, mute, solo, arm, meters and up to 4 sends. Knobs control volume, and the knob screens show the name or the dB value.
- **Device page:** 8 parameters of the selected channel's plugin or any mixer effect, in banks.
- **Transport:** play, stop, record, tap tempo, metronome, undo/redo, tempo entry and song position.

2. Requirements

What	Details
Akai Force	With Live Control mode (the Ableton controller mode), connected to your network by Wi-Fi or a USB-to-Ethernet adapter.
Akai Network Driver	Free, from the inMusic Software Center (<i>My Hardware</i> tab) or akaipro.com. Available for Windows and Intel Macs only.
FL Studio	With Python MIDI scripting (FL Studio 20.7 or newer). Developed and tested on FL Studio 2026 (v26.1).
Computer	Windows 10/11 (tested). macOS on Intel should work but hasn't been tested. Apple Silicon Macs can't run Akai's network driver.
Python	Not needed for the script (FL Studio has its own). Only the optional developer tools use it.

One device at a time. If several Forces or MPCs are on the network, only one can be in Live Control with the computer at once. Close Ableton Live too, so it doesn't compete for the port.

3. Installation

Step 1: Akai Network Driver and pairing

1. Install **Akai Network Driver** (inMusic Software Center → My Hardware, or akaipro.com). Restart the computer.
2. On the Force, press **MENU**, tap the **gear** icon (Preferences), open **Wi-Fi** and join the same network as the computer. Or use a USB-to-Ethernet adapter and enable **Ethernet** in Preferences.
3. Open the **Akai Network Driver** app and select your Force (IP address and serial number) under **Configured Remote Device**. If it isn't listed, use **Add a Device** and type its IP address. On the Force, hold **SHIFT** and tap **Info** in the Wi-Fi menu to see it.

After this, Windows has two new MIDI ports: `Akai Network - DAW Control` (used by this script) and `Akai Network - MIDI`.

Step 2: Install the script

Pick whichever is easiest:

A. Release download (recommended)

1. Download **AkaiForceLive-v1.0.0.zip** from the *Releases* page: `github.com/Pyrodrifter/akai-force-fl-studio/releases`
2. Unzip it and double-click **install.bat**.

B. One line in PowerShell

```
irm https://raw.githubusercontent.com/Pyrodrifter/akai-force-fl-studio/main/install.ps1 | iex
```

C. macOS (Intel, untested)

```
curl -fsSL https://raw.githubusercontent.com/Pyrodrifter/akai-force-fl-studio/main/install.sh | sh
```

The installer copies two files into FL Studio's user *Hardware* folder (it reads FL's own user-data location, normally `Documents\Image-Line\FL Studio\Settings\Hardware\Akai Force Live`), then checks that FL Studio and the Akai Network driver are present and prints the next steps.

Manual install: create the folder above and copy `device_AkaiForce.py` and `force_protocol.py` into it. That's all the installer does.

Step 3: Set up FL Studio

1. Open **Options** → **MIDI Settings**. If FL was already running, click **Update MIDI scripts**.
2. Under **Input**, select `Akai Network - DAW Control`, then set:

Setting	Value
Enable	On
Controller type	Akai Force (Live Control)
Port	Any free number, e.g. 1

3. Under **Output**, select `Akai Network - DAW Control` and give it the **same Port number**. This links the script's messages back to the Force; without it the Force stays on *waiting for communication*.
4. Leave `Akai Network - MIDI` alone; this script doesn't use it.

Step 4: Connect the Force

On the Force press `MENU` and tap **LIVE CONTROL**. Within a second or two the screen shows your mixer tracks and the knob screens flash **PERFORM**. You're connected.

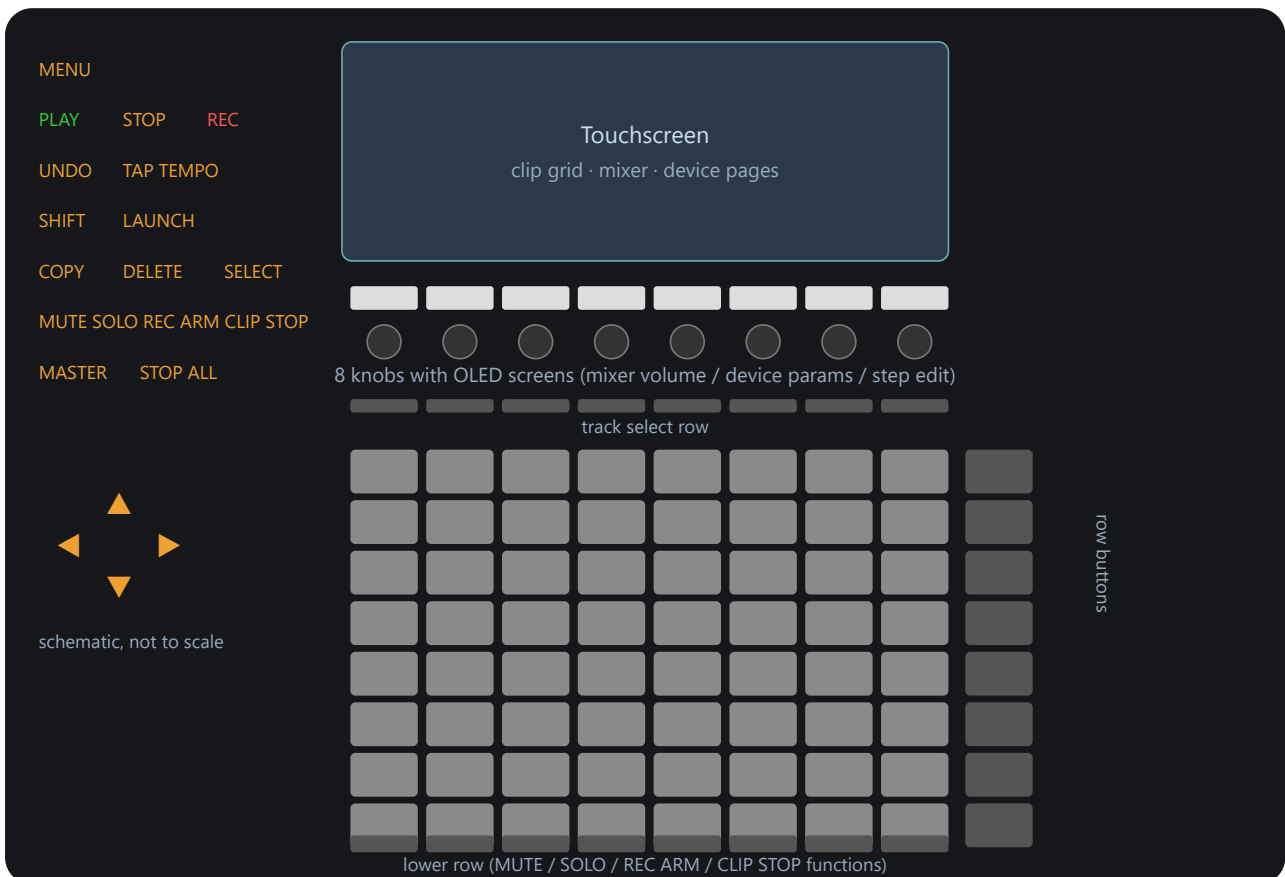
Check it: in FL, **View** → **Script output**, tab *Akai Network - DAW Control*, should read `Akai Force script v1.0.0 loaded` followed by `Akai Force connected`.

Updating and uninstalling

Update: run the installer again, then in FL click **Reload script** in Script output (or restart FL). **Uninstall:** `install.bat -Uninstall` (or `./install.sh --uninstall`), then set the `Akai Network - DAW Control` input back to (*generic controller*).

4. Quick start

1. Connect as above. The knob screens flash the current pad mode.
2. Press **LAUNCH** to cycle pad modes: PERFORM → PATTERNS → STEPS → KEYS → DRUMS → CHANNELS → SONG. **SHIFT** + **LAUNCH** goes back.
3. Turn a knob to change a mixer insert's volume; touch it to see the value in dB.
4. In **STEPS**, tap pads to program a beat on the current pattern, then press **PLAY**.
5. In **SONG**, tap any bar to jump there while the song plays, which is handy for chopping and rearranging live.



The controls referred to in this manual.

5. Controls

Hardware buttons

Control	Action
LAUNCH	Next pad mode (the mode name flashes on the knob screens). SHIFT+LAUNCH: previous mode.
PLAY / STOP / REC	FL transport. PLAY lights while playing.
TAP TEMPO	Tap tempo. SHIFT+TAP TEMPO: metronome on/off.
UNDO	Undo. SHIFT+UNDO: redo.
◀▶	Move the 8-track mixer window by one insert (SHIFT: by 8). In PERFORM they scroll blocks and in STEPS they page the steps. There, SHIFT+◀▶ moves the mixer.
▲▼	Scroll the pad rows of the current mode (playlist tracks, pattern pages, channels, octaves, drum banks, song pages). SHIFT scrolls further, or by a semitone in KEYS.
Track select row	Selects that mixer insert in FL (the selected one is lit white).
MUTE / SOLO / REC ARM / CLIP STOP	Choose what the lower button row does for the 8 visible inserts. In PERFORM they also choose what the row buttons on the right do.
SHIFT + first lower button	Quantize the selected channel.
COPY	Clone the current pattern.
SHIFT+DELETE	In STEPS: clear the selected channel's steps in this pattern.
SHIFT+SELECT	Pad pressure → aftertouch on/off (SELECT lights when on). For KEYS and DRUMS.
MASTER	Select the master track.
STOP ALL	PERFORM: stop all clips. Other modes: stop the transport.

Touchscreen

Use the icons on the left of the Force's screen to switch between its **clip grid**, **mixer** and **device** pages. The script fills them all:

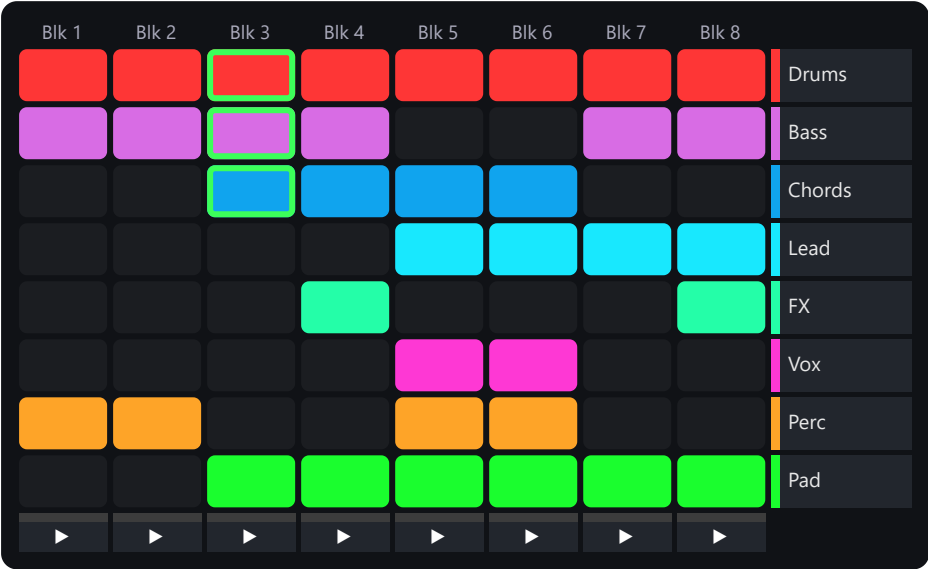
Screen element	In FL Studio
Clip grid	Mirrors the pads of the current mode, with names. Tapping a cell = pressing that pad.
Track headers	Mixer insert names and colours.
Faders, pan, mute, solo, arm	The 8 visible mixer inserts.
Sends A-D	Send levels to your send/FX inserts (see Mixer).
Tempo	Shows FL's tempo; type a new value to change it.

Position encoder	Moves the song position a beat per tick (SHIFT: one step).
Metronome / loop buttons	Metronome on/off; loop button switches pattern/song mode.
Quantize / Delete	Same as the hardware functions above.
Device page	8 plugin parameters with names and values, bank and device buttons.

6. Pad modes

The mode applies to the 8×8 pads, the clip grid on screen, and the row buttons to the right of the pads. The mixer, knobs and transport work the same in every mode.

PERFORM: clip launching

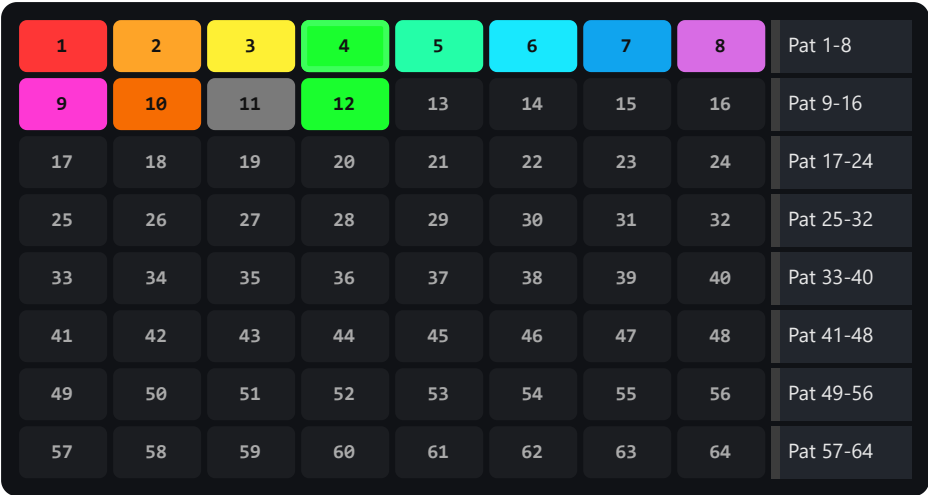


PERFORM: rows are playlist tracks, columns are Performance-Mode blocks. Pulsing = playing.

Control	Action
Pad	Launch that block's clip on that playlist track (other clips on the track stop). Release follows FL's trigger mode.
SHIFT + pad	Launch the whole column (every track's clip in that block), like an Ableton scene.
Row buttons (right)	Mute, solo or stop that playlist track, depending on the MUTE / SOLO / CLIP STOP button.
Lower row, CLIP STOP mode	Launch the block column above the button.
▲▼ / ◀▶	Scroll playlist tracks / blocks. FL outlines the visible area in the playlist.

PERFORM needs FL's **Performance Mode** switched on (Playlist → Performance mode). If it's off, the screen says *Turn on Perf Mode* and the pads do nothing.

PATTERNS



PATTERNS: one pad per pattern. The current pattern pulses.

Tap a pad to select that pattern (an empty slot creates it). ▲▼ pages through 8 patterns at a time (SHIFT: 64).

STEPS: step sequencer



STEPS: rows are channels, columns are steps. Lit = step on. The button row under the pads shows the playhead.

Control	Action
Tap a pad	Toggle that step. Lit = on, dark = off, exactly as in FL's channel rack.
Hold a pad + turn knobs	Edit that step: knob 1 pitch, 2 velocity, 3 release, 4 fine pitch, 5 pan, 6 Mod X, 7 Mod Y. The knob screens show the values while you hold. Editing never switches the step off.
SHIFT + pad / row button	Select that channel.
◀▶	Page through the pattern 8 steps at a time (up to its real length).
▲▼	Scroll channels (SHIFT: 8 at a time).
SHIFT+DELETE	Clear the selected channel's steps.

Channels whose notes were written in the **piano roll** are labelled (*piano roll*) and show no steps, just like FL's channel rack. The script treats a synth plugin whose notes use more than one pitch as piano roll. If that hides a real step pattern, set `STEPS_HIDE_PIANO_ROLL = False` (see Settings).

KEYS: melodic playing

B5	C6	C#6	D6	D#6	E6	F6	F#6	Chromatic
F#5	G5	G#5	A5	A#5	B5	C6	C#6	Major
C#5	D5	D#5	E5	F5	F#5	G5	G#5	Minor
G#4	A4	A#4	B4	C5	C#5	D5	D#5	Dorian
D#4	E4	F4	F#4	G4	G#4	A4	A#4	Mixolydian
A#3	B3	C4	C#4	D4	D#4	E4	F4	Harm minor
F3	F#3	G3	G#3	A3	A#3	B3	C4	Maj pent
C3	C#3	D3	D#3	E3	F3	F#3	G3	Min pent

KEYS (Chromatic): rows are a 4th apart, bottom-left is C3. Root notes in blue. The right-hand buttons choose the scale.

Plays the **selected channel**, and records like any MIDI keyboard. The pads are velocity sensitive.

Control	Action
Row buttons (right)	Pick the scale: Chromatic, Major, Minor, Dorian, Mixolydian, Harmonic minor, Major pentatonic, Minor pentatonic. In a scale, every pad is in key and rows are 3 scale degrees (a 4th) apart.
▲▼	Octave up / down. SHIFT+▲▼ moves the root by a semitone (the bottom-left pad is the root).
Colours	Root notes use the channel's colour (deep blue if it has none), naturals are grey and sharps darker.

DRUMS

E5	F5	F#5	G5	C8	C#8	D8	D#8
C5	C#5	D5	D#5	G#7	A7	A#7	B7
G#4	A4	A#4	B4	E7	F7	F#7	G7
E4	F4	F#4	G4	C7	C#7	D7	D#7
C4	C#4	D4	D#4	G#6	A6	A#6	B6
G#3	A3	A#3	B3	E6	F6	F#6	G6
E3	F3	F#3	G3	C6	C#6	D6	D#6
C3	C#3	D3	D#3	G#5	A5	A#5	B5

DRUMS: four 4x4 banks of 16 notes. Bottom-left bank starts at C3 (MIDI 36), the standard FPC layout.

For FPC, Slicex and drum plugins on the selected channel. ▲▼ moves by a bank of 16 notes.

CHANNELS



CHANNELS: one pad per channel-rack channel. Tap to play, SHIFT+tap to select.

Each pad triggers one channel-rack channel. Perfect for kits built from separate samples. SHIFT+pad selects the channel (the selected one blinks). ▲▼ pages by 8.

SONG



SONG: one pad per bar. Marker sections take the marker's colour; the current bar pulses.

Each pad is one bar of the song. Tap to jump there. The current bar pulses and the page follows playback. Entering SONG switches FL to song mode. With **playlist markers**, each marker's bar shows its name and the section takes its colour. Without markers, 4-bar phrases alternate shades. ▲▼ pages by a row (SHIFT: 64 bars).

FL can't tell scripts where markers are, so the script finds them when you enter SONG mode (only while stopped), briefly moving the playhead and putting it back.

7. Mixer, knobs and screens

Feature	Details
Mixer window	8 consecutive mixer inserts, moved with ◀▶ (SHIFT: 8). FL outlines them in the mixer.
Knobs (mixer)	Volume of the 8 inserts. Hold SHIFT for fine control. The Force chooses whether knobs are on the mixer or device page (its KNOBS button / screen pages).
Knob screens	Show the insert name, the dB value while you touch the knob, and a volume bar.
Meters	Left/right peak meters for each strip on the mixer page.
Mute / solo / arm	On screen, or with the lower button row in MUTE / SOLO / REC ARM mode.
Sends A-D	Route each strip to up to 4 send inserts. By default these are the first inserts whose names contain <i>send</i> , <i>reverb</i> , <i>verb</i> , <i>delay</i> , <i>bus</i> or <i>fx</i> . Or list them in <code>SEND_TRACKS</code> . Turning a send on an unrouted strip creates the route.

8. Device page (plugins)

Control	Action
Knobs (device)	8 parameters of the current plugin. The knob screens show names, or values while touched.
Prev / next bank	Next 8 parameters (the bank number shows as <i>Bank 2/12</i>).
Prev / next device	Select the previous/next channel, or the previous/next effect slot in FX-chain mode.
SHIFT + prev/next device	Switch between the selected channel's plugin and the selected mixer insert's effect chain.
Focused effect window	If an effect's window is focused in FL, the page follows it automatically.

Only named parameters are listed (VST plugins report thousands of empty slots).

9. Colours

Everything on the Force takes its colour from FL: channel colours (STEPS, CHANNELS, KEYS roots), mixer insert colours (track headers, track buttons), playlist block and track colours (PERFORM) and pattern colours (PATTERNS). The Force has a fixed 70-colour palette, so each FL colour is matched to the nearest one.

Give your channels real colours. FL's default channel grey can only become a pale grey-white on the pads. Bright, saturated colours (red, orange, cyan, green, purple...) look best. Colour each mixer insert like the channel that feeds it and a sound keeps its colour across every mode.

Dark colours are brightened automatically so pads never look unlit (`MIN_BRIGHTNESS`).

10. Settings reference

Advanced options live at the top of `device_AkaiForce.py` in FL's Hardware folder. Edit them in any text editor, then click **Reload script** in View → Script output. Re-running the installer overwrites them.

Setting	Default	What it does
PAD_AFTERTOUCH	False	Start with pad pressure → aftertouch on (also toggled live with SHIFT+SELECT).
SEND_TRACKS	()	Mixer inserts for send knobs A-D, e.g. (20, 21). Empty = find them by name.
SEND_KEYWORDS	send, reverb, ...	Words that mark an insert as a send when auto-detecting.
STEPS_HIDE_PIANO_ROLL	True	Hide piano-roll channels' notes in STEPS.
MIN_BRIGHTNESS	200	Colours darker than this (0-255) are brightened.
SCALES	8 scales	The scales offered in KEYS (name + semitones).
VOLUME_STEP / PARAM_STEP	0.004 / 0.008	How far a knob tick moves volume / plugin parameters.
FINE	0.2	SHIFT multiplier for knobs.
FLASH_TIME	1.2 s	How long the mode name stays on the knob screens.
PING_INTERVAL / PONG_TIMEOUT	1 s / 8 s	Connection keep-alive and how long before it counts as lost.
HEAL_INTERVAL	0.25 s	Background refresh of one pad row, repairing any dropped network message.
FX_SLOTS / MAX_PARAM_SCAN	10 / 1024	Effect slots per insert / plugin parameters scanned for names.

11. Troubleshooting

Symptom	Fix
Force says <i>waiting for communication</i>	Check MIDI Settings: the DAW Control input uses the Akai Force script and is enabled, and the DAW Control output has the same Port number . Make sure Ableton isn't open and the Force is paired in the Akai Network Driver app.
Script isn't in the Controller type list	Click Update MIDI scripts in MIDI Settings or restart FL. Check that <code>Settings\Hardware\Akai Force Live</code> contains both .py files.
Script output: <i>Output port not linked</i>	Give the DAW Control output the same Port number as the input.
No <i>Akai Network</i> ports in FL	Install the Akai Network Driver, restart, pair the Force, and check both are on the same network.
PERFORM pads do nothing	Turn on Performance Mode in the playlist (the screen says <i>Turn on Perf Mode</i>).
Pads look white or pale	Those channels/tracks have FL's default grey. Give them colours (chapter 9).
A channel shows (<i>piano roll</i>) in STEPS	Its notes come from the piano roll. Play it in KEYS, or set <code>STEPS_HIDE_PIANO_ROLL = False</code> .
Hint bar: <i>FL refused that just now</i>	FL was busy (e.g. loading a project). Press again.
Brief disconnect while loading big projects	Normal. It reconnects by itself within a few seconds.
Two Forces/MPCs on the network	Only one can use Live Control with the computer at a time.
Apple Silicon Mac	Not supported: Akai's network driver is Windows / Intel-Mac only.

12. For developers

File	Purpose
device_AkaiForce.py	The FL Studio controller script.
force_protocol.py	The Force's Live Control protocol: constants, message builders and the full control map. No dependencies.
fl_sim.py	Runs the script outside FL against a fake project. <code>python fl_sim.py</code> runs the automated checks. <code>--live</code> drives a real Force. Needs <code>pip install mido python-rtmidi</code> .
force_probe.py	Talks to the Force with no DAW: handshake, message decoder, demo layout.
install.ps1 / install.bat / install.sh	Installers.
docs/build_manual.py	Builds this manual (HTML → PDF with Chrome or Edge).

Protocol summary

The protocol was worked out from Ableton Live's own *Akai_Force_MPC* remote script and verified on hardware. All SysEx is `F0 47 00 <product> <type> ... F7` (Force product id `0x40`).

Message	Format
Ping (host)	<code>F0 47 00 7F 00 F7</code> , about every 1-3 s
Pong (Force)	<code>F0 47 00 40 01 F7</code> . The Force also sends it on its own while waiting.
Text	<code>F0 47 00 40 10 id0 id1 lenHi lenLo ascii... F7</code> , e.g. track name <code>(0, t)</code> , clip name <code>(0, 16+row*8+col)</code> , knob screen <code>(18, 16+k)</code> , tempo <code>(3, 0)</code>
Pads	Channel 13 notes 16-79 (<code>16+row*8+col</code>); colour CC 40-103 on the same layout; the note velocity sets the clip state (0 empty, 2 stopped, 3 queued, 4 playing, 7 recording)
Pad pressure	Repeated note-ons while held: the first is the velocity, later ones are pressure, 0 = release
Knobs	Channel 14 CC 0-7 (mixer page) / 8-15 (device page), relative two's complement; touch = notes 0-15
Screen mixer	Channel 2-9 (one per strip): CC 0 volume, 1 pan, 3-6 sends, 124/125 meters; notes 0 solo, 1 mute, 5 arm
Colours	Values 8-77 = Live's 70-colour palette; 0-7 and 127 are state colours

The complete map is in `force_protocol.py`.

13. License and credits

Released under the **MIT License**. See `LICENSE` in the repository.

Akai, Akai Professional, Force and MPC are trademarks of inMusic Brands. FL Studio is a trademark of Image-Line. Ableton and Live are trademarks of Ableton AG. This project is independent and not affiliated with or endorsed by any of them.

Source, issues and updates: `github.com/Pyrodrifter/akai-force-fl-studio`